4.3 Left: **The Stoner–Anderson *equation of state* for white-dwarf matter, that is, the relationship between the matter's density and its resistance to compression. Plotted horizontally is the density to which the matter has been squeezed. Plotted vertically is its resistance (the percentage increase of pressure that accompanies a 1percent increase of density). Along the curve is marked the squeeze pressure (equal to internal pressure), in multiples of the pressure of the Earth's atmosphere. *Right:* The circumferences (plotted horizontally) and masses (plotted vertically) of white-dwarf stars as computed by Chandrasekhar using Eddington's Braunschweiger mechanical calculator. Along the curve is marked the density of the matter at the center of the star, in grams per cubic centimeter.**

For stars with low or extremely high central density, which Chandrasekhar had studied on the steamer, he had found the solution to the differential equation and the resulting stellar structures in Eddington'sbook; but for stars with intermediate densities Eddington's book was of no help and, despite great effort, Chandrasekhar was not able to deduce the solution using mathematical formulas. The mathematics was too complicated. There was no recourse but to solve his differential equation numerically, on a computer.

Now, the computers of 1934 were very different from those of the 1990s. They were more like the simplest of pocket calculators: They could only multiply two numbers at a time, and the user had to enter those numbers by hand, then turn a crank. The crank set into motion a complicated morass of gears and wheels which performed the multiplication and gave the answer.

Such computers were precious machines; it was very hard to gain access to one. But Arthur Eddington owned one, a "Braunschweiger" about the size of an early 1990s desk-top personal computer; so Chandrasekhar, who by now had become well acquainted with the great man, went to Eddington and asked to borrow it. Eddington at the time was embroiled in a controversy with Milne over white dwarfs and was eager to see the full details of white-dwarf structure worked out, so he let Chandrasekhar cart the Braunschweiger off to the rooms in Trinity College where Chandrasekhar was living.

The calculations were long and tedious. Each evening after dinner Eddington, who was a fellow of Trinity College, would ascend the stairs to Chandrasekhar's rooms to see how they were coming and to give him encouragement.

At last, after many days, Chandrasekhar finished. He had met Ambartsumian's challenge. For each of ten representative white-dwarf stars, he had computed the internal structure, and then from it the star's total